

02LQD Specification and Simulation of Digital Systems

2018/02/09

1. Combinational circuit design (5 points)

Thermometer-to-binary (TM2B) encoders are important components of Flash Analog-to-Digital Converters (FADC). The thermometer outputs a pure binary signal X on 7 bits. The encoder receives X and generates Z on 3 bits according to the following table:

X	Z
0000000	000
0000001	001
0000011	010
0000111	011
0001111	100
0011111	101
0111111	110
1111111	111

Design in VHDL an encoder implementing the above specification according to the following description styles:

1. behavioural using IF THEN ELSEIF statements
2. behavioural using the CASE WHEN statement.

2. FSM design (10 points)

In digital communication, a special synchronization pattern, known as a *preamble*, is used to indicate the beginning of a packet. For example, the Ethernet II preamble includes eight repeating octets of "10101010". Design a synchronous circuit that generates the "10101010" pattern. The circuit has a 1 bit input signal, *start*, and a 1 bit output signal, *data_out*. When *start* is '1', the "10101010" output will be generated in the next eight clock cycles. Reset is synchronous.

1. Draw the state diagram (2 points)
2. Design the Moore FSM resorting to the VHDL behavioural description style with 2 processes (3 points)
3. Design the HLSM (3 points).

3. Memory-based design (15 points)

Given a $2^3 \times 4$ ROM, design a system that counts how many ROM values are at 0. It should start counting when a *Go* signal is asserted and it should set a *Finish* signal to 1 when counting is over and the result is available on *Counter*. The system acts as top-level entity and its structure is composed of a ROM and of an interface module. The interface module is a FSM that reads data sequentially from the ROM, compares them to 0 and updates the count accordingly. Reset is synchronous. Define the signals exchanged by the interface module and the ROM. Create a suitable testbench.

Exam rules: upload on Portale della didattica in section Elaborati:

- running VHDL code and testbench
- short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Monday February 12 12.00 am

- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.